



CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics

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Problem

Current Situation

- Cognitive assessment is commonly conducted using traditional tests.
- Traditional assessments may cause stress, anxiety, and reduced motivation.

Existing Issues

- Limited user engagement.
- Lack of behavioral analytics.
- Results are often limited to scores.

Supporting Evidence

- Recent studies support game-based cognitive assessment as a more engaging approach.



Research Gap

Existing Studies and Systems

- Game-based cognitive assessments effectively evaluate memory, attention, reasoning, and problem-solving skills.
- Adaptive gameplay improves user engagement and motivation.
- Artificial Intelligence and Machine Learning enhance personalization and assessment accuracy.
- Existing platforms such as Lumosity, Peak, Elevate, NeuroNation, and CogniFit provide cognitive training and performance monitoring.



Research Gap

Identified Gaps

- Most systems focus on only one or a few cognitive domains.
- Limited behavioral analytics based on gameplay interactions.
- Assessment results are often limited to numerical scores.
- Few systems implement Dynamic Difficulty Adjustment (DDA).
- Limited use of Machine Learning for cognitive classification.
- Lack of Cognitive Archetype identification.
- No unified platform combines assessment, adaptive gameplay, AI analytics, and cognitive profiling.



Proposed solution

CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics

- Integrates multiple cognitive domains:
 - Memory
 - Attention
 - Reasoning
 - Reflexes
 - Executive Strategy
- Implements Dynamic Difficulty Adjustment (DDA).
- Utilizes Machine Learning-based Cognitive Archetype Classification.
- Provides behavioral analytics and performance visualization.
- Delivers a comprehensive and engaging cognitive assessment experience.



Objectives

To develop an Adaptive Cognitive Training Game with AI-driven Skill Evaluation and Behavioral Analytics, the purpose is to build a web-based game platform which will make use of AI technology, machine learning, and dynamic difficulty adjustment for cognitive skill assessment and behavior analytics.

Specific Objectives

Specifically, this study aims to:

1. To develop four cognitive games which will test for:
 - Spatial/Visual Memory
 - Logical/Mathematical Reasoning
 - Reflexes and Concentration
 - Strategic Thinking
1. To implement a DDA (Dynamic Difficulty Adjustment) system that automatically adjusts the level of difficulty in relation to the level of the players' proficiency.
2. To Create a game data tracking system which tracks reaction time, accuracy, decision-making patterns, and cognitive performance.
3. To implement a machine learning model for classification called Random Forest Classifier from Scikit-Learn using a Python Flask backend to classify users into different Cognitive Archetypes.



Objectives

Specific Objectives

5. To Create a web application to track cognitive performance using interactive analytics dashboard created using React.js & [Chart.js](#).

6. Provide users with personalized cognitive insights and performance reports through AI-driven behavioral analytics.

7. Evaluate the developed system using the **Game Experience Questionnaire (GEQ)** in terms of:

- Competence
- Sensory and Imaginative Immersion
- Flow
- Tension and Annoyance
- Challenge
- Positive Affect
- Negative Affect

8. Evaluate the effectiveness of application using pretest and posttest



Scope

The study focuses on:

- Web-based cognitive training platform.
- User registration and authentication.
- Cognitive mini-games.
- Dynamic Difficulty Adjustment.
- Cognitive Archetype Classification.
- Behavioral Analytics Dashboard.
- Performance Tracking and Visualization.

Technologies Used

- React.js
- Phaser.io
- Python Flask
- Scikit-Learn
- MySQL Database

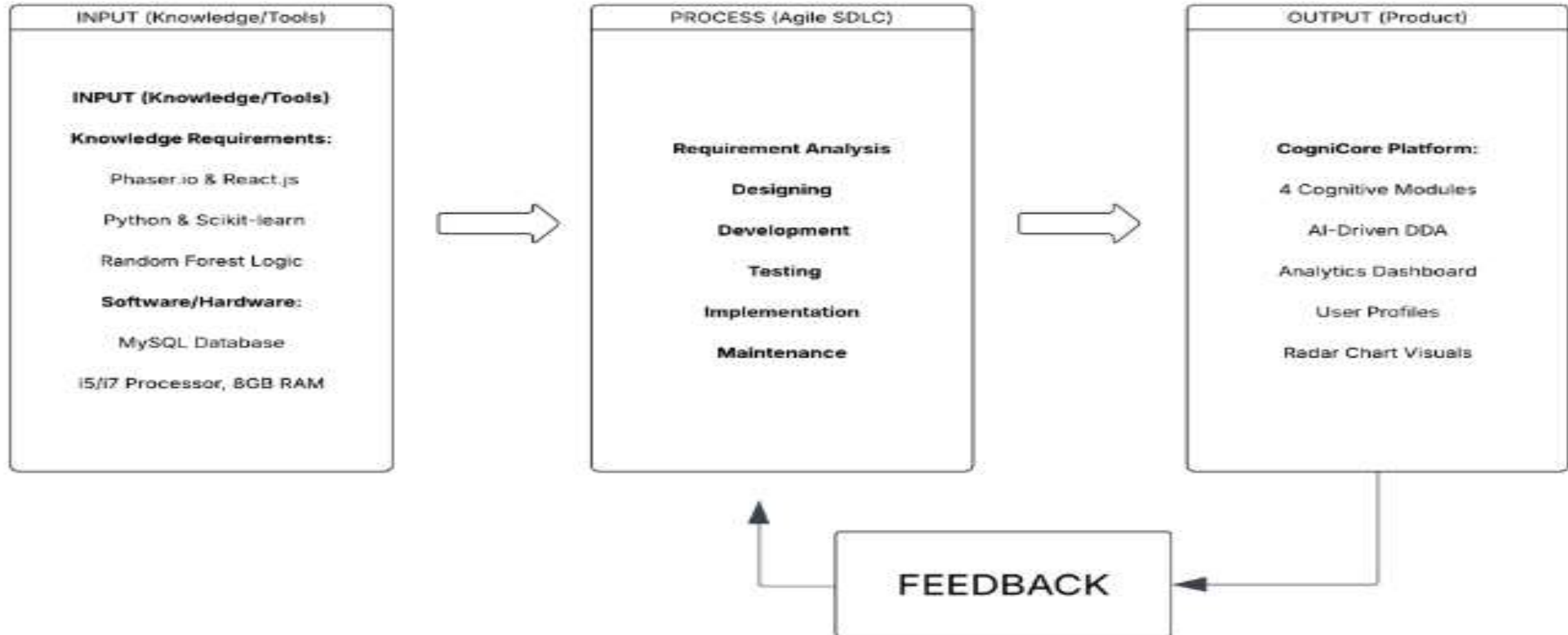


Limitation

- Not intended for medical diagnosis.
- Requires stable internet connection.
- Limited to developed cognitive modules.
- Browser-based accessibility only.
- Cognitive assessment depends on gameplay interactions.
- Optimized primarily for English-speaking users.



IPO





SDLC

Agile Software Development Methodology

Planning

- Identify project goals and requirements.
- Gather user needs through preliminary investigation.

Design

- Create system architecture, UI designs, and game mechanics.
- Design cognitive assessment modules and database structure.

Development

- Develop cognitive mini-games using Phaser.io.
- Implement AI-driven cognitive classification and analytics.
- Integrate Dynamic Difficulty Adjustment (DDA)

Testing

- Conduct functionality, usability, and performance testing.
- Identify and fix system errors and issues.

Deployment

- Deploy the web-based platform for user access and evaluation.

Review and Improvement

- Collect feedback from users and experts.
- Refine system features and improve overall performance.



Research Design

Developmental-Descriptive Research Design

Developmental Research

- Focuses on the design and development of CogniCore.
- Covers the development of cognitive mini-games using Phaser.io.
- Implements AI-driven cognitive classification and analytics.
- Integrates Dynamic Difficulty Adjustment (DDA).

Descriptive Research

- Evaluates the developed system's quality and user experience.
- Measures system acceptability using ISO 25010.
- Assesses player experience using the Game Experience Questionnaire (GEQ).
- Collects and analyzes feedback from respondents and experts.



Respondents

A. Focus Group (20 Participants)

- Selected participants who will complete the full training program.
- Participate in Pre-test and Post-test activities.
- Used to evaluate training effectiveness.
- Data used for AI learning curve calibration.

B. Evaluation Group (100 Respondents)

- Composed of students and experienced gamers.
- Evaluate system usability and engagement.
- Complete ISO 25010 and GEQ questionnaires.
- Provide feedback on overall user experience.

C. Technical Experts (10 IT Professionals)

- Serve as technical validators.
- Evaluate system functionality and performance.
- Assess the Random Forest AI model.
- Validate database security and software quality.

Sampling Technique

- Purposive Sampling



Instruments

- **ISO 25010 Software Quality Questionnaire:** Used to evaluate functionality, usability, and reliability.
- **Game Experience Questionnaire (GEQ):** Measures player immersion, flow, and psychological affect.
- **Expert Validation Rubric:** For technical and psychological validation by the 10 IT and SME validators.
- **Pre-test / Post-test Assessment:** To establish a cognitive baseline and measure training effectiveness for the focus group.



Conceptual Design

- **Presentation Layer:** Built with Phaser.io (Gameplay) and React.js (User Interface).
- **Logic Layer:** A Python/Flask backend running a Random Forest Classifier to analyze behavioral data.
- **Data Layer:** A MySQL database for secure storage of user profiles and gameplay telemetry.
- **Real-time DDA Loop:** A feedback mechanism that adjusts difficulty every 10 seconds to maintain the "Flow State."



Game Play/ Mechanics (Game Only)

- **Module 1: Memory:** Spatial-visual pattern matching tasks.
- **Module 2: Logic:** Mathematical reasoning and algorithmic puzzles.
- **Module 3: Reflex:** High-speed target tracking and focus assessment.
- **Module 4: Strategy:** Executive pathfinding and resource management.
- **Scoring Logic:** Points are calculated based on reaction time (hesitation_ms) and accuracy.
- **DDA Thresholds:** Difficulty increases at >90% accuracy and decreases at <60% accuracy.



Story Board (Game Only)

- ❖ **Scene 1: Entry & Authentication**
- ❖ User registration and profile creation via a secure React.js portal.
- ❖ **Scene 2: The Game Nexus (Hub)**
- ❖ Central menu for selecting the 4 Cognitive Modules (Memory, Logic, Reflex, Strategy).
- ❖ **Scene 3: Active Play & DDA Loop**
- ❖ Phaser.io gameplay featuring real-time scaling of difficulty based on user accuracy.
- ❖ **Scene 4: The Analytics Lab**
- ❖ Interactive dashboard displaying the AI-generated Archetype and 360° Radar Charts.



COGNICORE STORY BOARD (GAME ONLY)

A Cognitive Training Game for Cognitive Assessment and Archetype Classification





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CERTIFIED
ISO 9001
ISO 21001



College of Computer Studies

System Demonstration(15 mins)